

challenging the cosmological principle

angular homogeneity scale in the era of large galaxy surveys

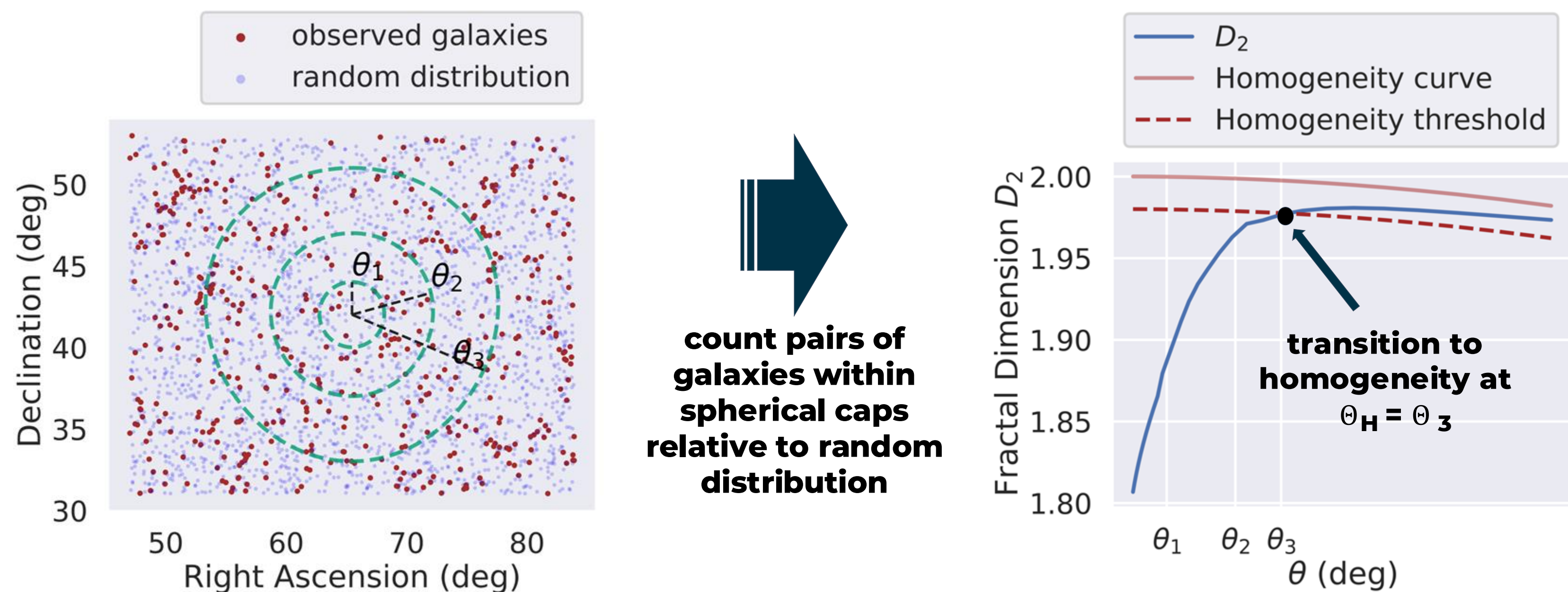
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intent

- To build a consistent, cosmological model-independent computational method to determine the **angular homogeneity scale** θ_H as well as its associated **uncertainty**
- To determine the minimum area needed from future large-scale galaxy surveys to obtain trustworthy measurements of the angular homogeneity scales of our Universe

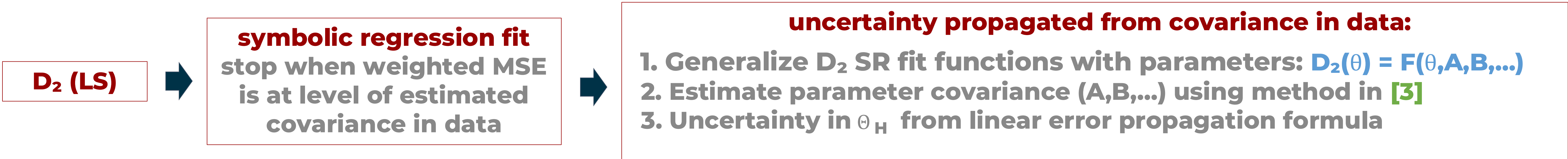
theory



Homogeneity threshold is defined as 1% away from D_2 for homogeneous region.

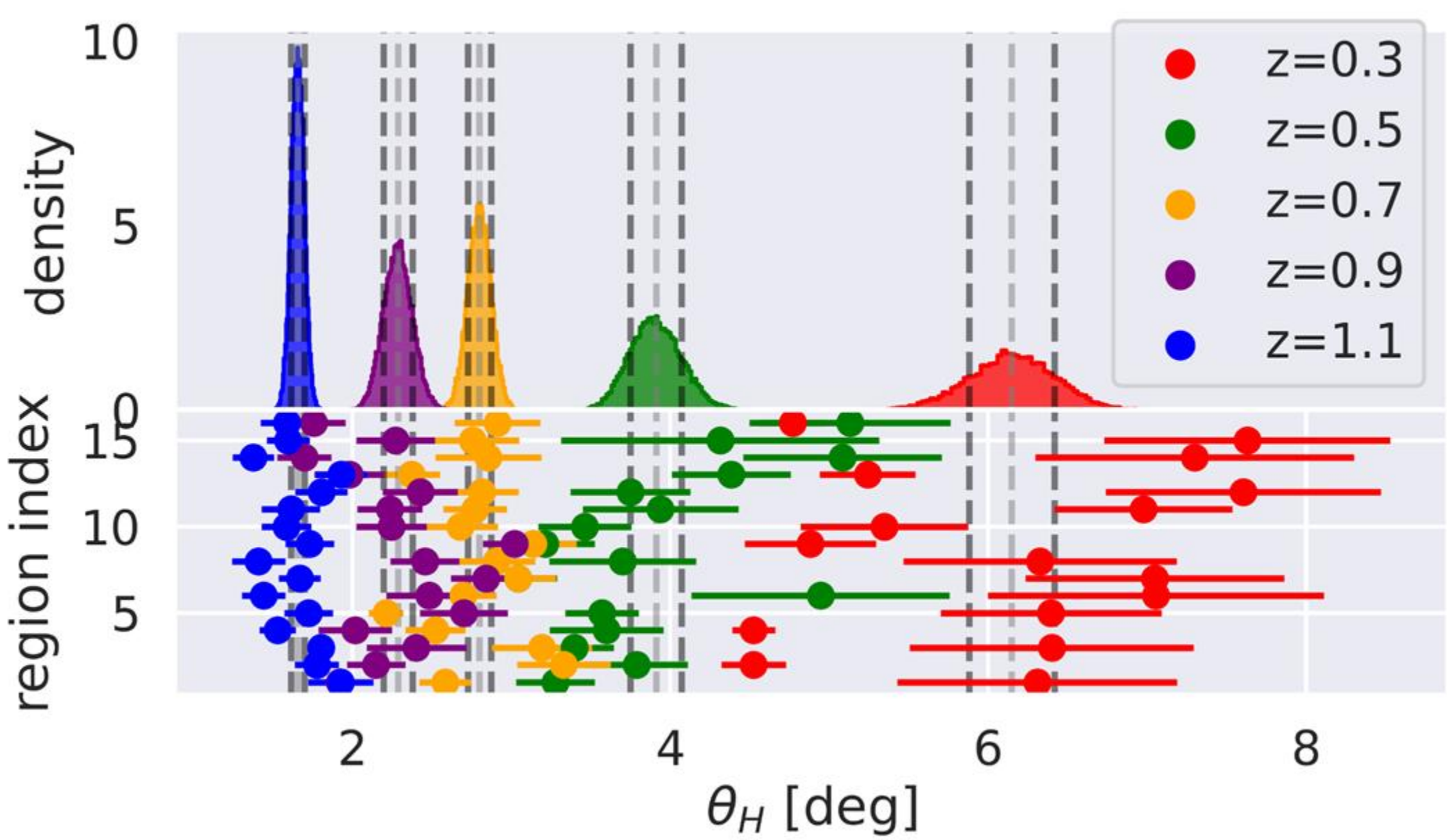
methodology

- Conducted a comprehensive literature review of the existing methods based on the **Landy-Szalay (LS)** estimators [1] for estimating θ_H .
- Using state-of-the-art numerical galaxy survey simulations, we implemented and tested both **parametric (pre-defined fit functions)** and **non-parametric methods powered by Symbolic Regression (SR)** - a machine learning technique that uncovers analytical expressions directly from data - to determine the fractal dimension profiles.
- We developed approaches to estimate the uncertainty associated with each method from covariance matrix estimation of the fractal dimension profiles in the data [4, 5].
- The diagram below shows the pipeline for one SR method.

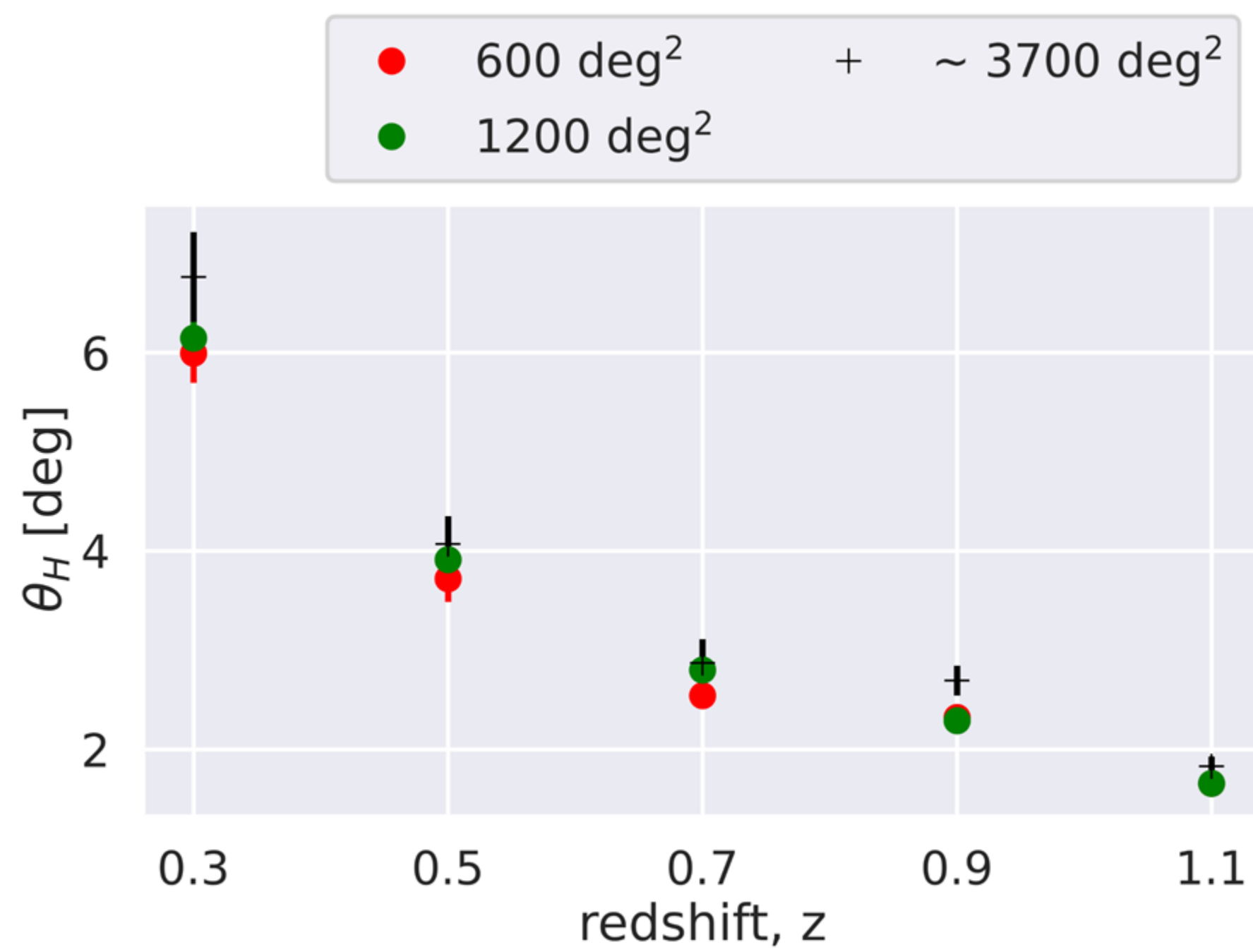


results

application to the MICECAT 2 simulation catalogue [2]
(magnitude cut in DES survey i-band ≤ 23.5)



Caption: θ_H from many 1200 deg² regions.



Caption: bootstrap mean from 16 1200 deg² and 16 600 deg² regions compared to full MICECAT within DES survey completeness region.

conclusions

- The new SR method effectively captures key trends in fractal dimension profiles and θ_H estimates, remaining consistent with traditional parametric approaches while offering greater interpretability and flexibility.
- Survey footprint strongly affects robustness: larger areas yield more accurate and less dispersed θ_H estimates for the DES i-band magnitude limit used here.
- At lower redshifts, θ_H estimates show large scatter, highlighting the need for more regions to obtain reliable bootstrap distributions.
- Current and upcoming surveys (e.g., Euclid), with deeper magnitude limits and wider sky coverage, will provide more reliable tests of cosmological homogeneity - a foundational pillar of the cosmological principle.

references

- [1] DOI: 10.48550/arXiv.1806.04541
- [2] DOI: 10.48550/arXiv.1312.2013
- [3] DOI: 10.1029/2005WR004804
- [4] GitHub: rmjarvis/TreeCorr
- [5] DOI: 10.1093/mnras/stac1458

The authors acknowledge support from FCT, under project UID/04434/2025.